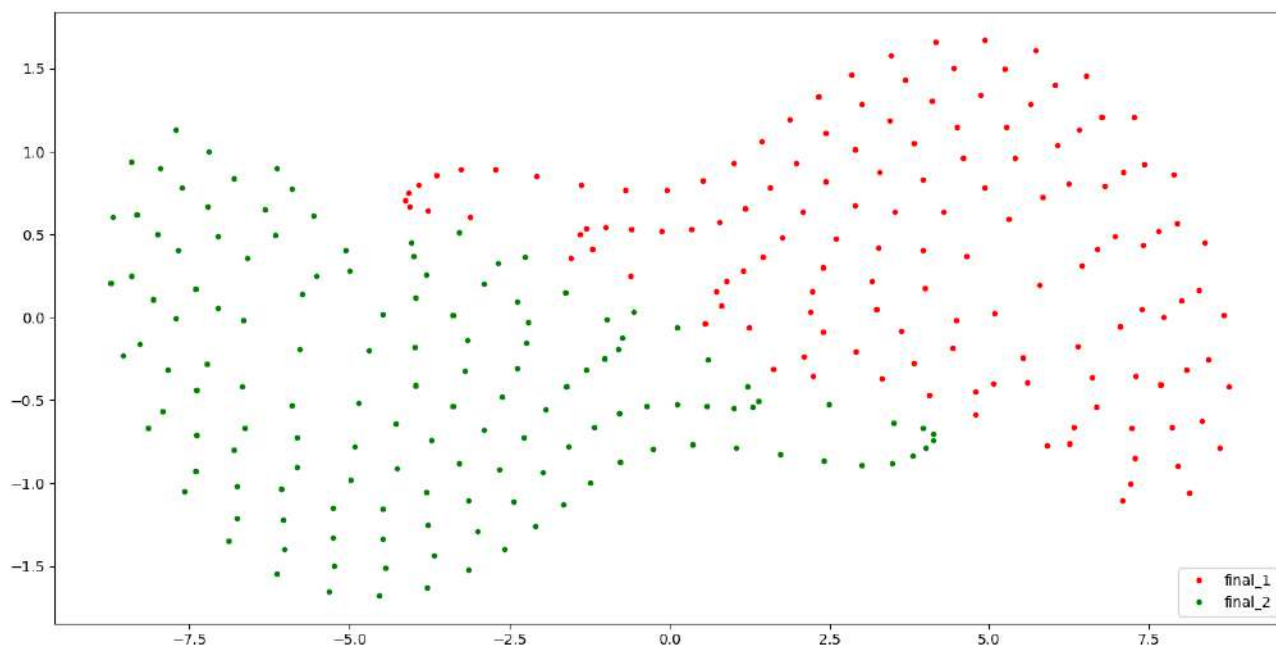
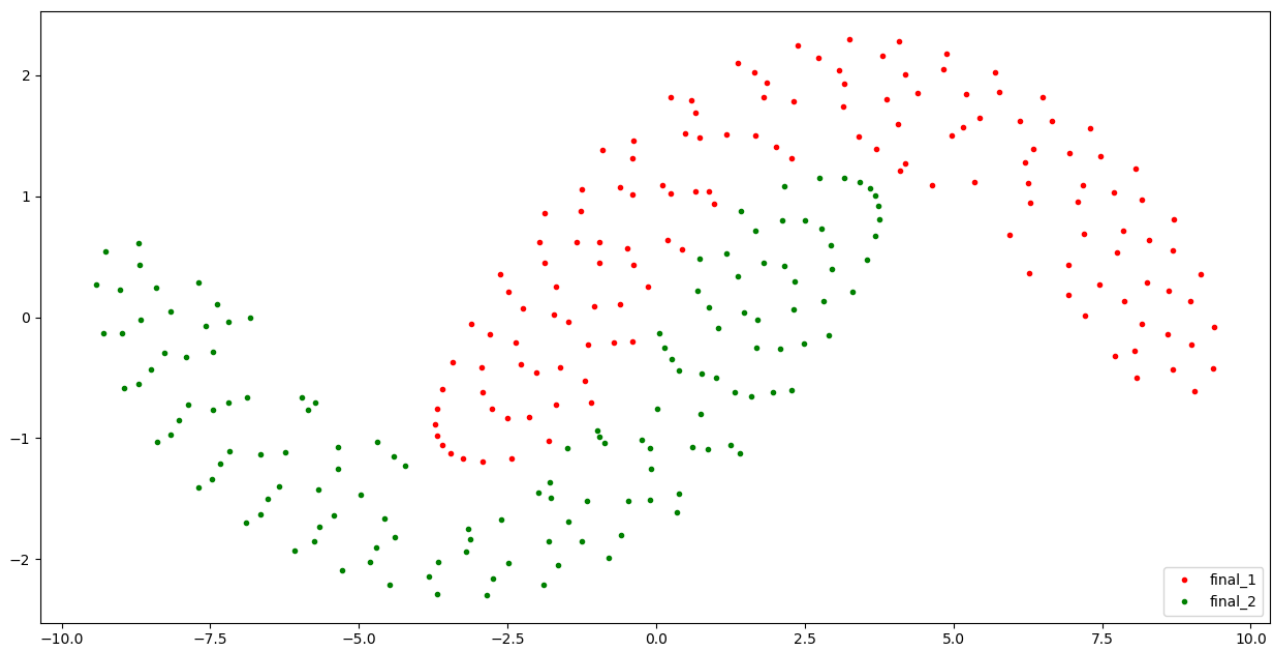
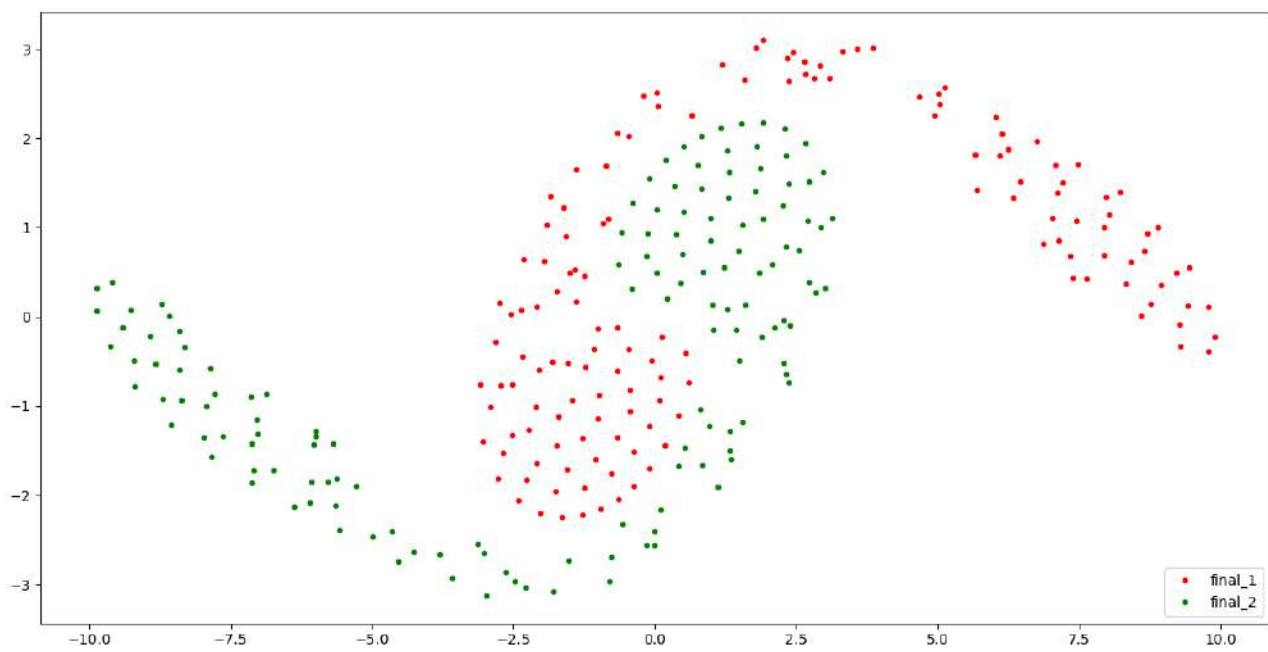
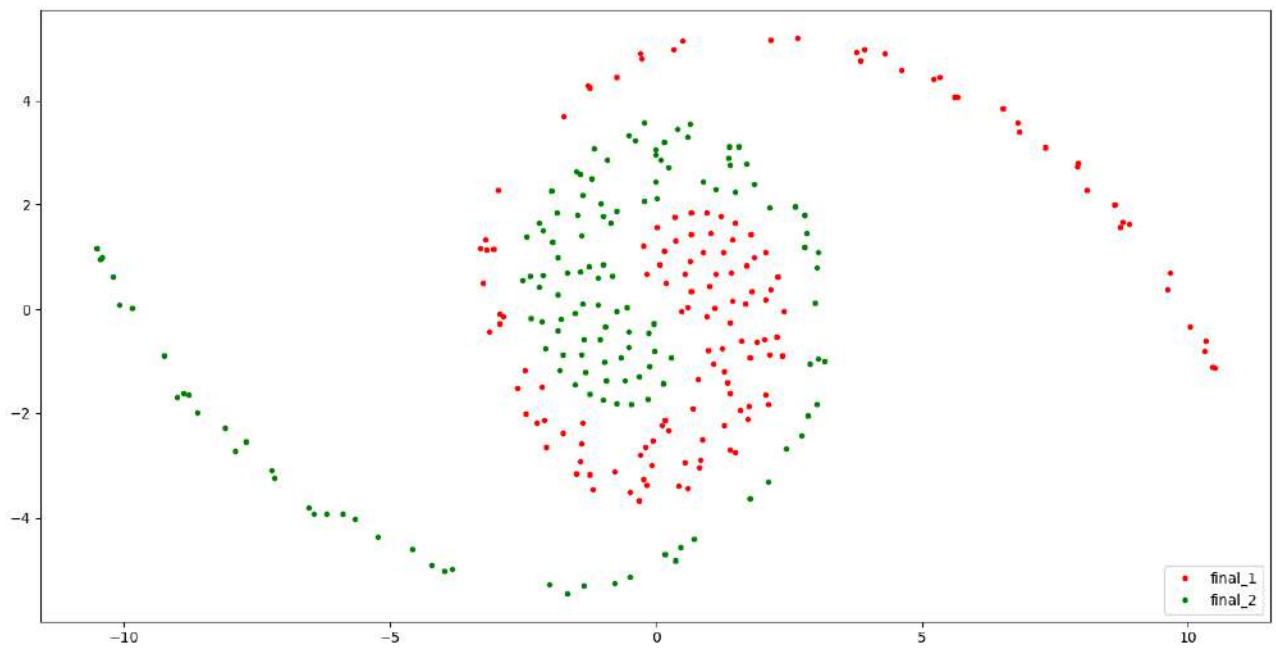
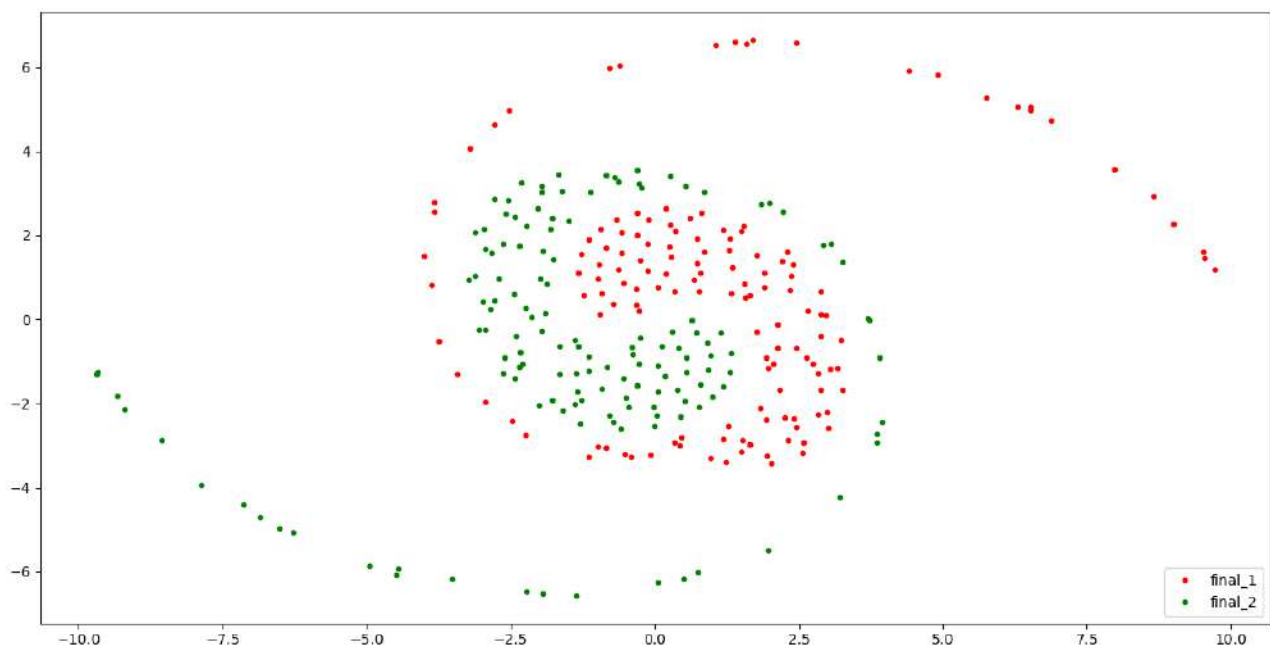


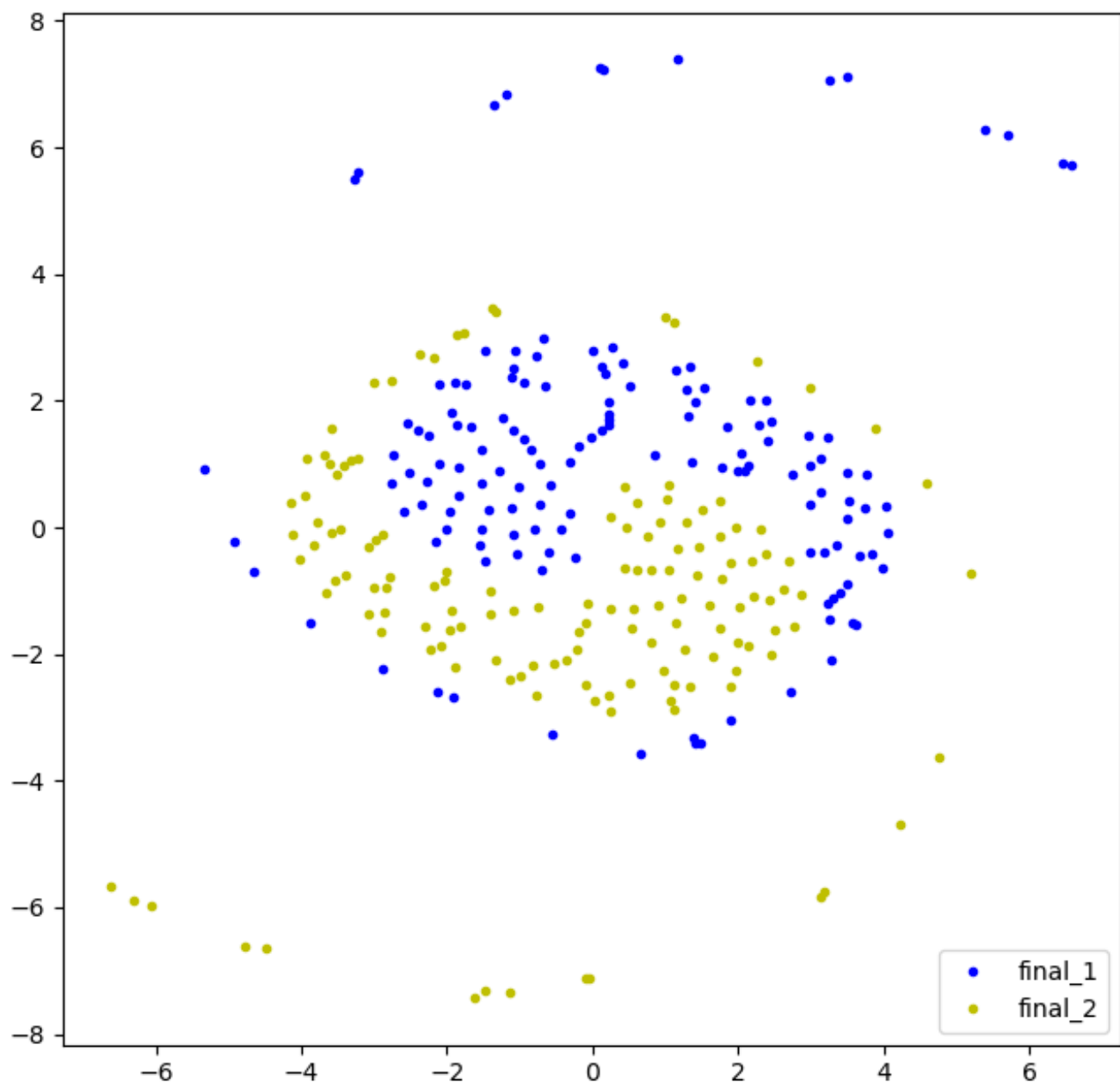


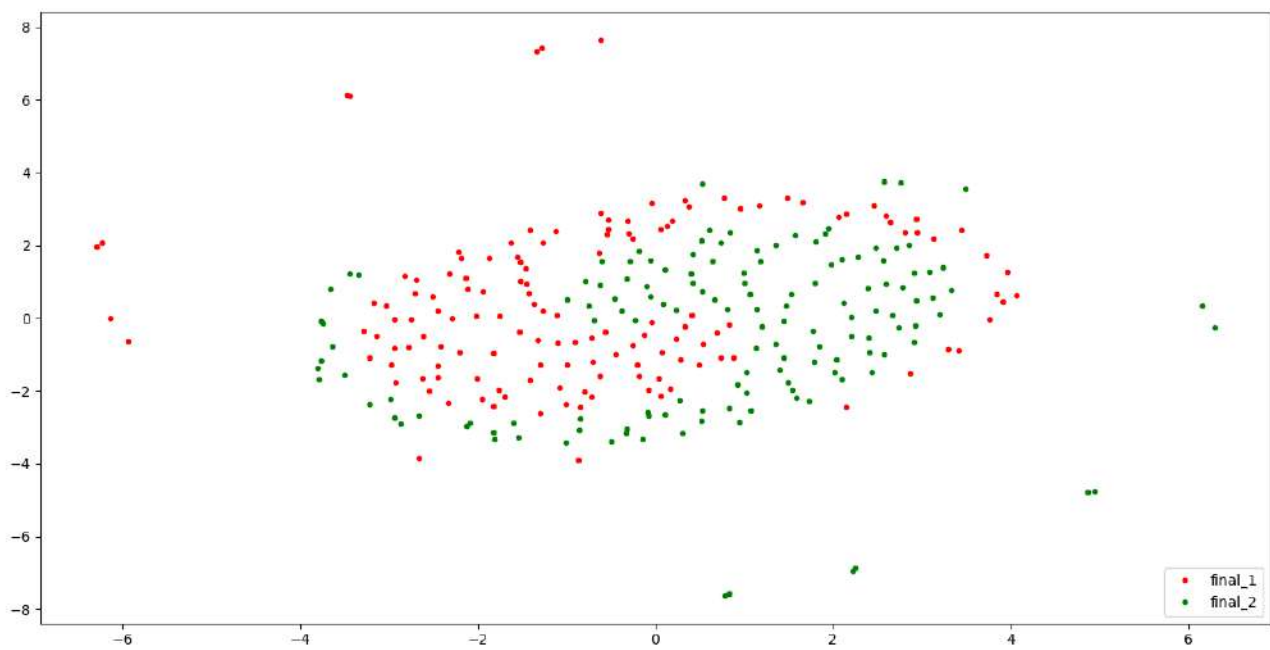


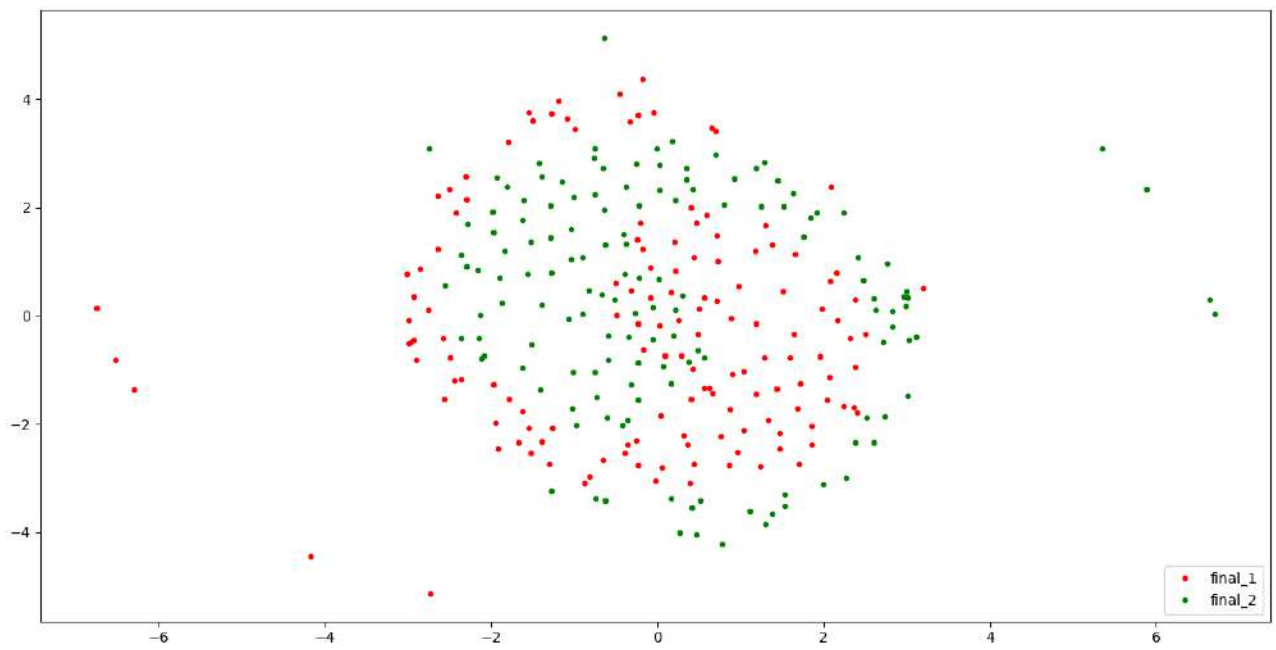


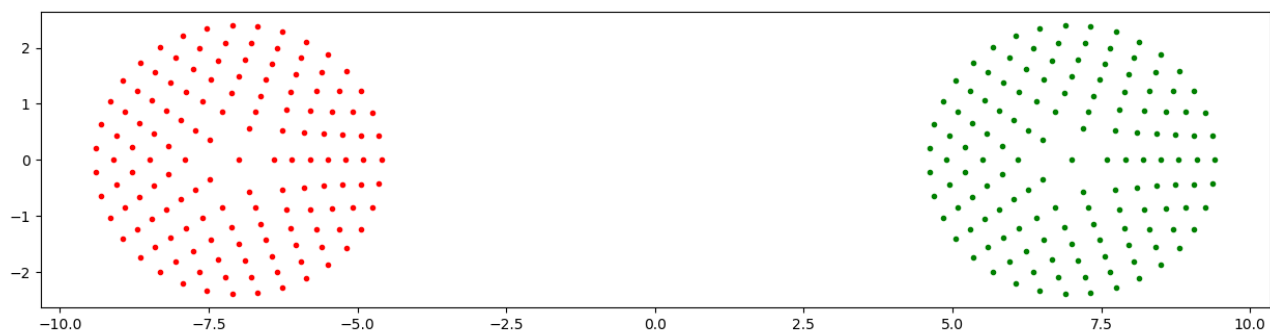


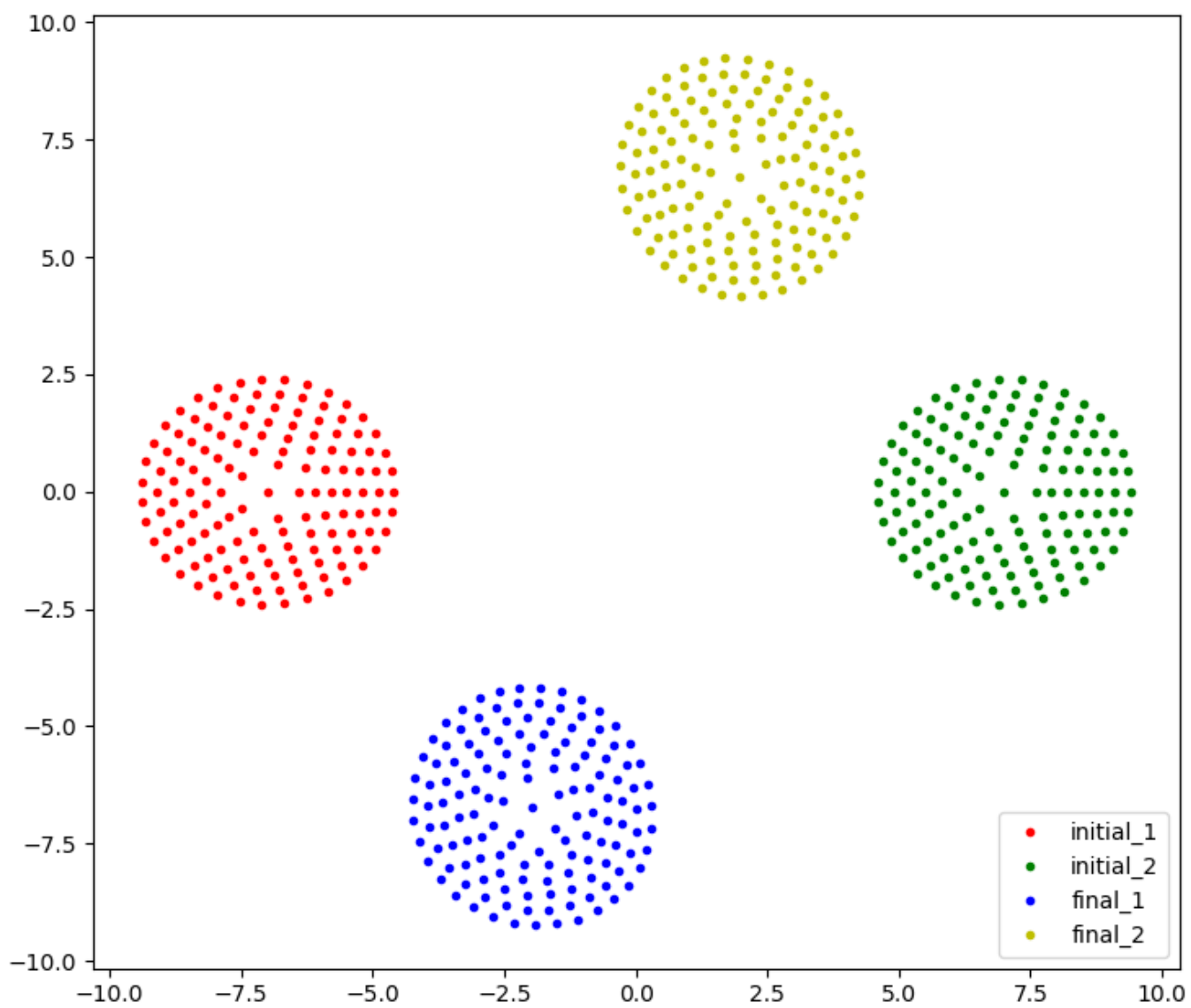


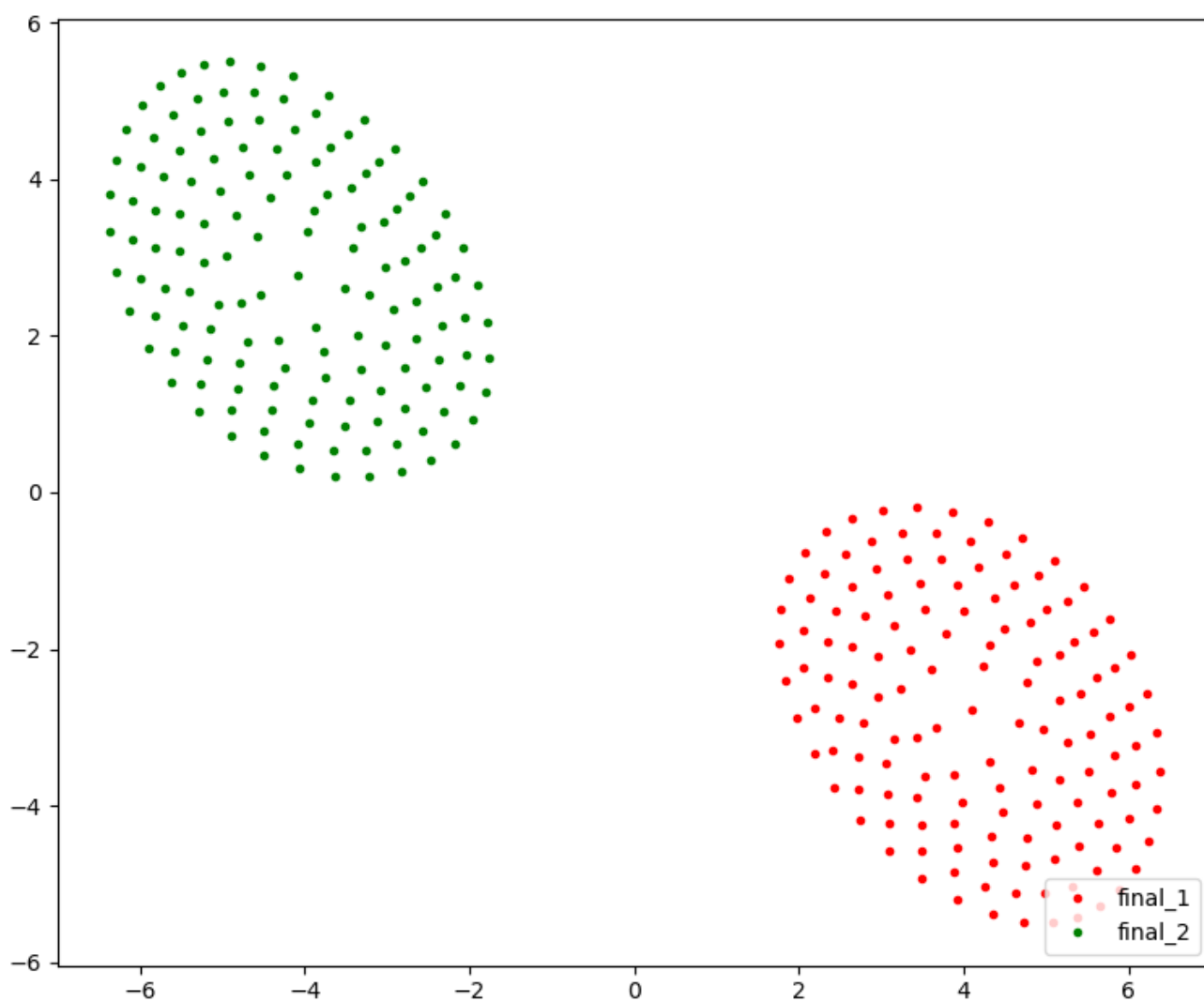


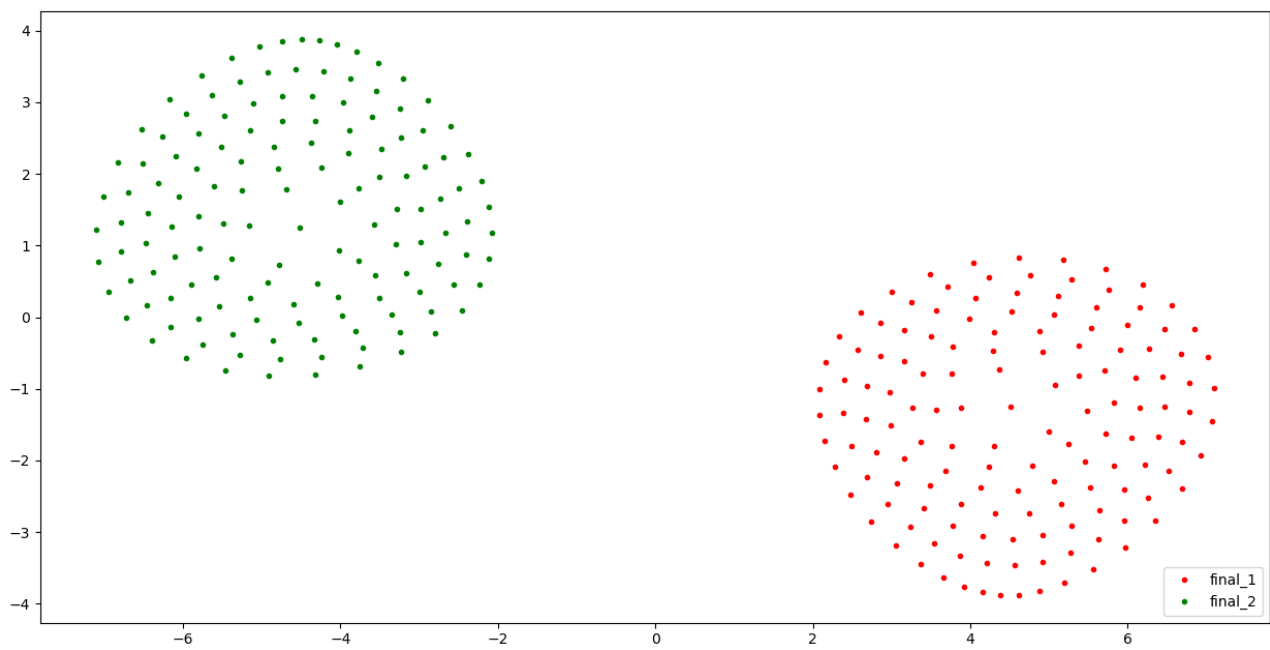


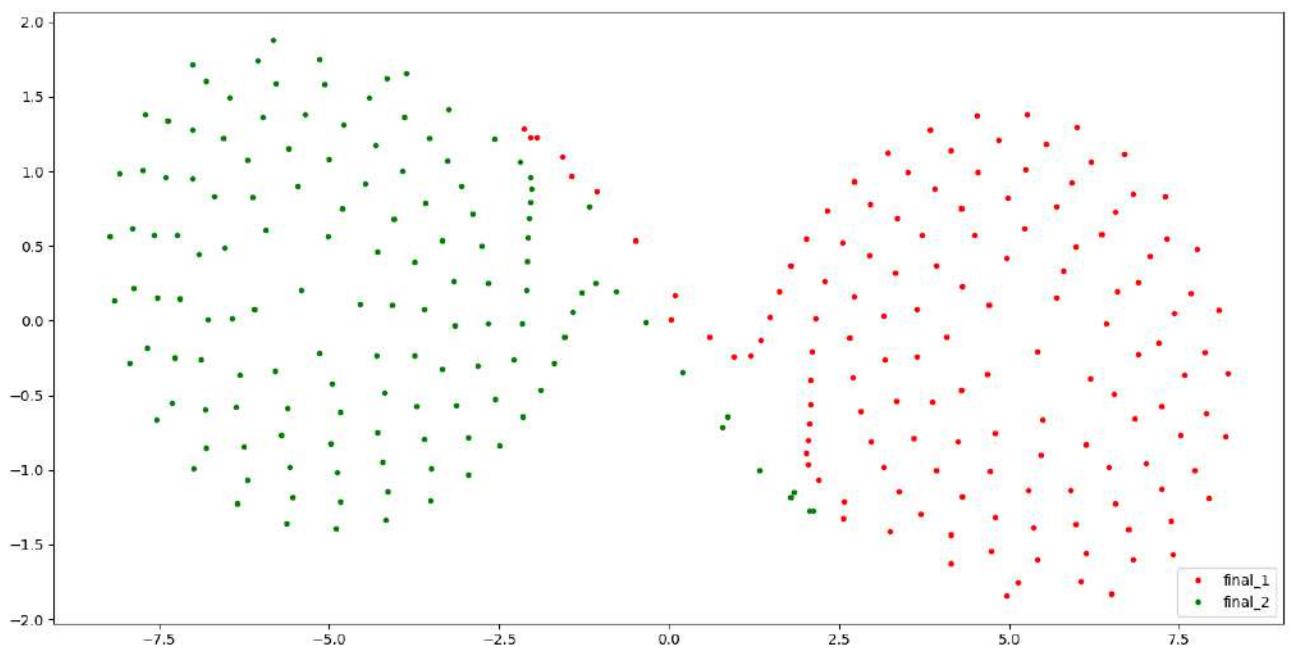


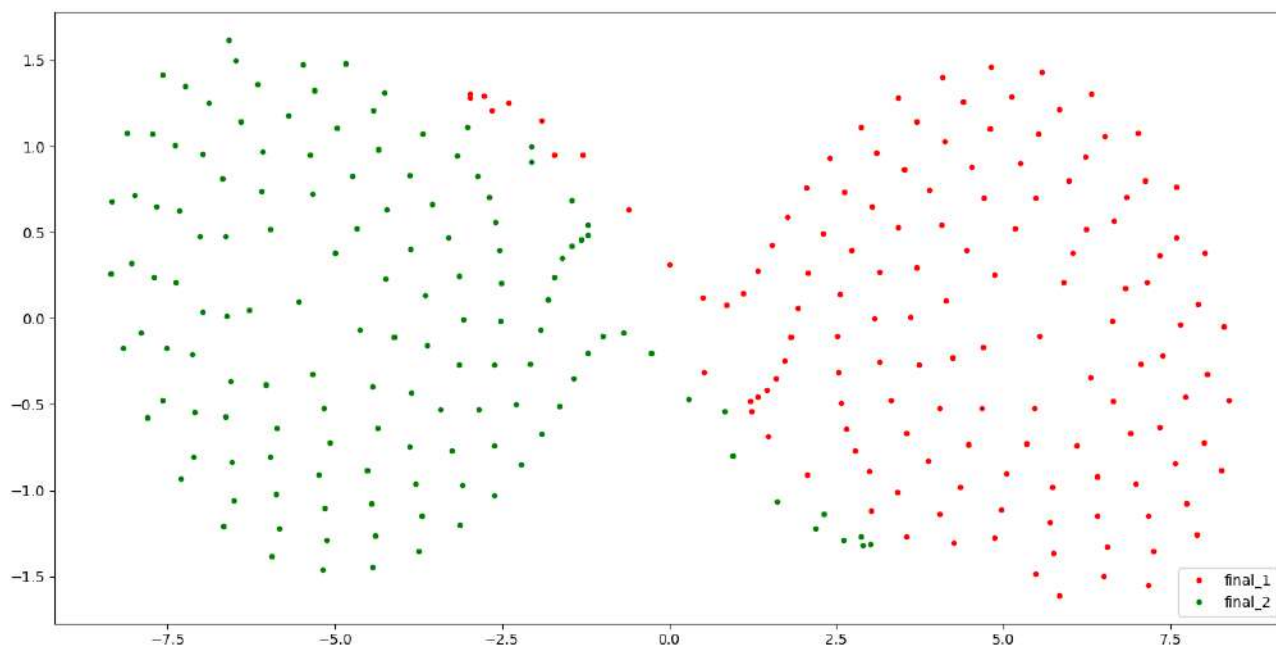


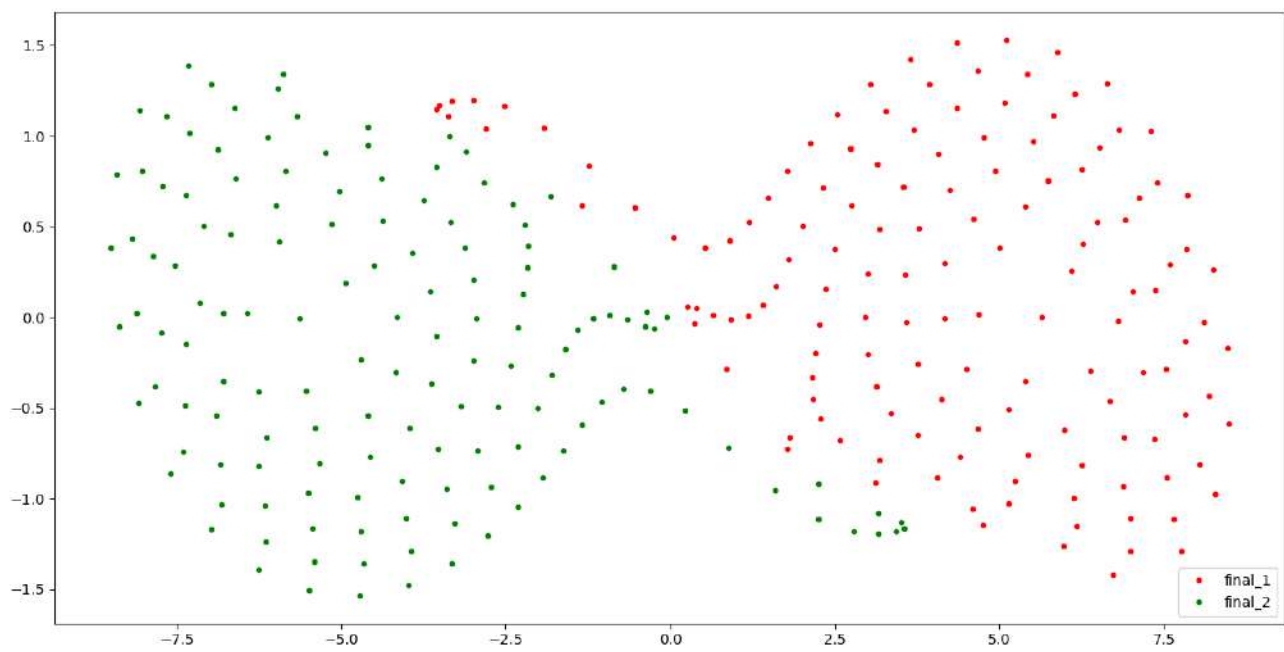












```

1  import numpy as np
2  import math as mh
3  import matplotlib.pyplot as plt
4  import numba
5
6  TOS=8
7  cente_r1= -4.68+0.0j
8  cente_r2= 4.68+0.0j
9  d=0.3
10 J=7
11 M=5
12 Tau1=100.0
13 del_t_x=4
14
15 def initial_config(Tau1,cente_r,d,J,M):
16     nop= int((J*(J+1))*M/2.0 + 1.0)
17     Pos= np.array([cente_r])
18     for j in range(J+1):
19         r_j=(j+1)*d
20         for i in range(j*M):
21             z_ij = cente_r + r_j*np.exp(1j*((i+1)*(2*np.pi)/(j*M)))
22             Pos=np.append(Pos,z_ij)
23     return nop,Pos,Tau1/nop * np.ones(nop),np.abs(Pos[-1]-Pos[-2])
24
25 numba.njit
26 def velocity_cal(i,j,Pos,Tau,delta_sq):
27     z_i=Pos[i]
28     z_j=Pos[j]
29     r_d=z_i-z_j
30     return np.complex(0,0.5/(mh.pi))*Tau[j]*r_d/((abs(r_d))**2+delta_sq)
31
32 numba.njit
33 def velocity(Pos,Tau,d_x,N):
34     Vel_i=np.zeros_like(Pos)
35     delta_sq=(del_t_x*d_x)**2
36     for i in range(N):
37         j=list(range(N))
38         j.remove(i)
39         j=np.array(j)
40         Vel_i[i]=np.sum(velocity_cal(i,j,Pos,Tau,delta_sq))
41     return Vel_i
42
43 numba.njit
44 def Vortex_simulation(d_t):
45     N,Pos,Tau,d_x=initial_config(Tau1,cente_r1,d,J,M)
46     N2,Pos2,Tau2,d_x=initial_config(Tau1,cente_r2,d,J,M)
47     Pos=np.append(Pos,Pos2)
48     Tau=np.append(Tau,Tau2)
49     N=N+N2
50     Vel=0j*np.zeros(N)
51     Path_tracker=np.array([Pos.copy()])
52     tim=0
53     while tim<TOS:
54         Pos_temp = Pos + 0.5*(velocity(Pos,Tau,d_x,N)*d_t)
55         Pos += velocity(Pos_temp,Tau,d_x,N)*d_t
56         tim += d_t
57         Path_tracker=np.append(Path_tracker,[Pos.copy()],axis=0)
58     return Path_tracker
59
60 def Path_plotting(d_t):
61     plt.figure(1)
62     nop,Pos,Tau,d_x=initial_config(Tau1,cente_r1,d,J,M)
63     #plt.plot(Pos.real,Pos.imag,'r.',label='initial_1')
64     nop,Pos,Tau,d_x=initial_config(Tau1,cente_r2,d,J,M)
65     #plt.plot(Pos.real,Pos.imag,'g.',label='initial_2')
66     Path_track=Vortex_simulation(d_t)
67     plt.plot(Path_track[-1][:nop].real,Path_track[-1][:nop].imag,'r.',label='final_1')

```



```
68 plt.plot(Path_track[-1][nop:].real, Path_track[-1][nop:].imag, 'g.', label='final_2')
69 plt.legend(loc='lower right')
70 plt.show()
71
72
```